

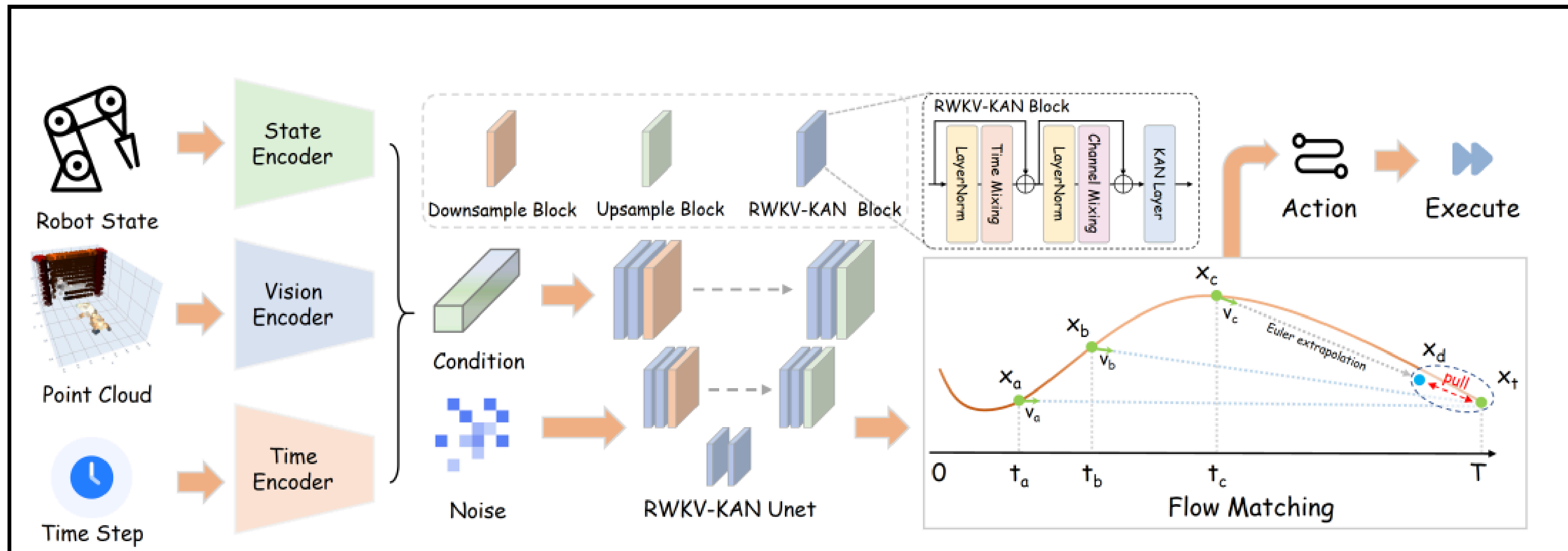
Weekly Progress Report

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Reference Work

- **KAN-We-Flow:** A high-performance, **single-task robotic policy** using RWKV, GroupKAN, and Flow Matching. It proved the primitives work, but only for isolated, non-linguistic skills.
- **The Problem**
- **Task Isolation & Linguistic Blindness:** Modern robotics requires general-purpose agents that can follow instructions and generalize across tasks. Standard specialized policies are "blind" to language and cannot scale to diverse multi-task environments.

KAN We Flow



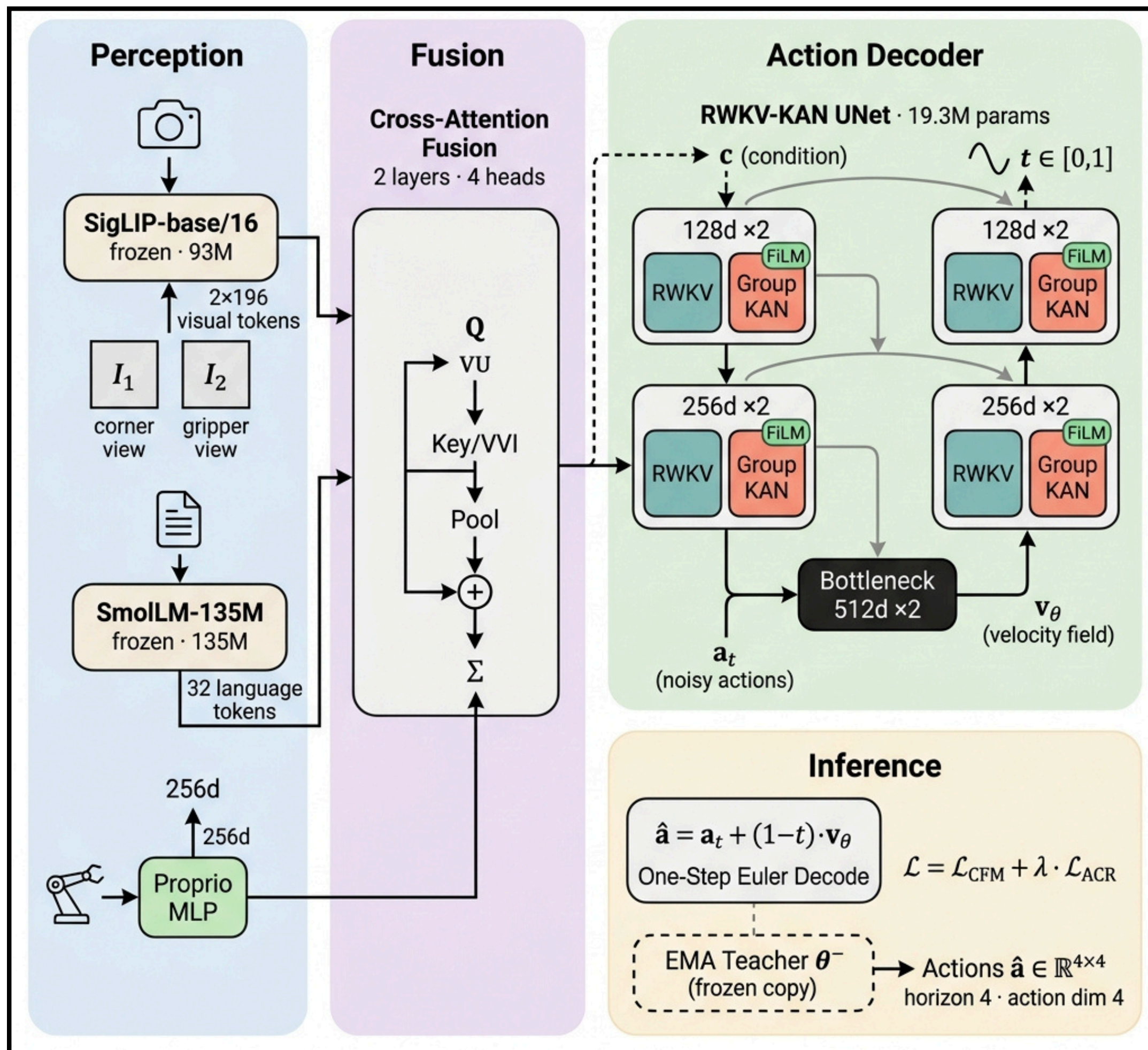
our Work

- **The Limitation**

Open-Loop, Single-Modal Execution: KAN-We-Flow lacks a language interface, requires one model per task, and has no internal "reliability check." It cannot detect or correct its own failures during execution.

- **Research Goal**

- **KANFlow-VLA:** To build the **Multi-Task, Language-Conditioned RWKV-KAN VLA**. We combine the efficiency of KAN-We-Flow with a multi-task MT-50 instruction-following backbone and a Reliability-Aware (RCAR) self-correction loop.



with very less retraining

After ablation assessment, changes applied to the architecture

10 episode per task -
total 100 episodes: **55-65%**
successrate on MT-10
and sometimes it go to 20%

Key Difference		
Aspect	KAN-We-Flow	KANFlow-VLA
Task setup	Single-task imitation	Multi-task VLA (MT-10/50)
Inputs	Vision + state	Vision + state + language
Language conditioning	No	Yes
Backbone	RWKV + GroupKAN + CFM	RWKV + GroupKAN + CFM
Encoders	Task-specific	SigLIP + SmolLM
Main contribution	Efficient policy backbone	Efficient multi-task VLA integration
Scale	Per-task policy	Unified 50-task policy
Expected SR1	71–92% - single-task imitation	50–55%
Research focus	Efficiency in imitation learning	Few-shot efficient VLA scaling